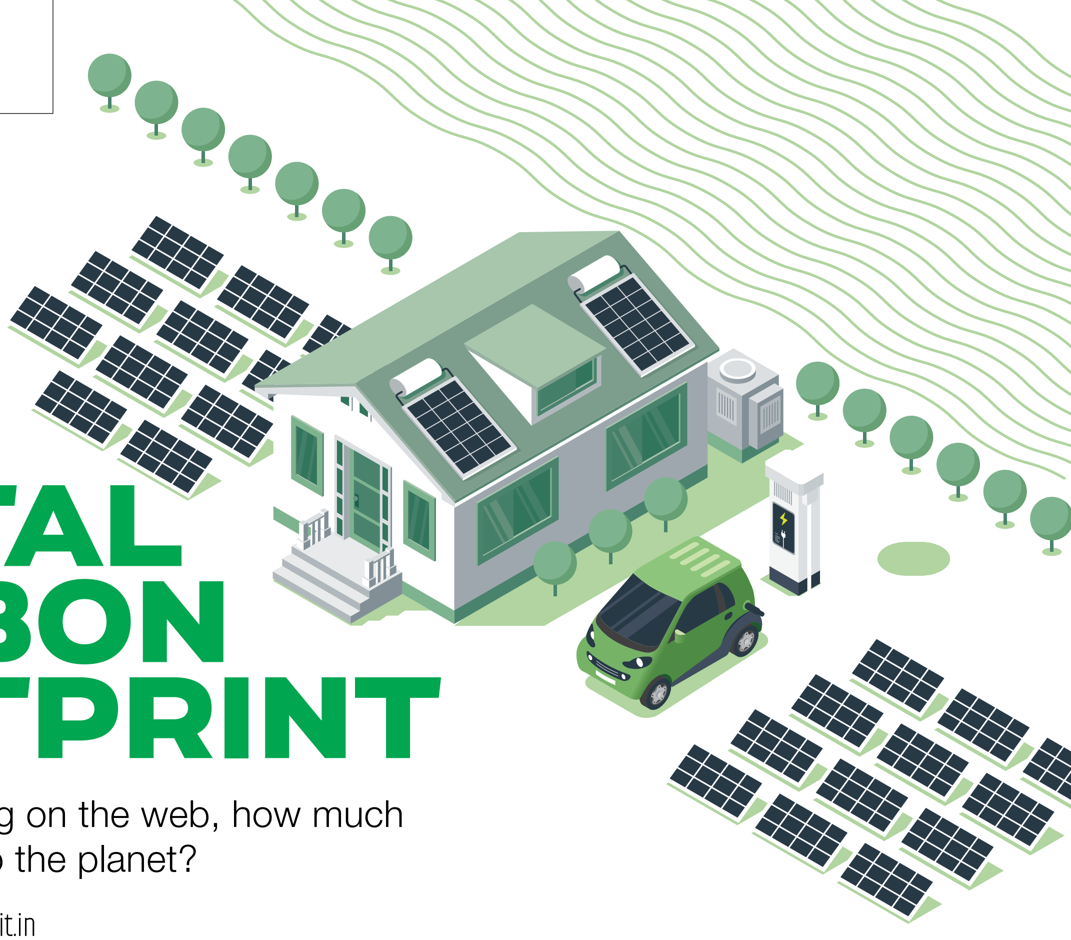

CUTTING DOWN DIGITAL CARBON FOOTPRINT

In the age of residing on the web, how much tax are we paying to the planet?

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The energy required for sustaining the digital world is ever growing. Every mail sent, every Google search made, every video streamed or every download initiated taps the internet data centres into action. And of course, the consumption on devices also draws massive amounts of energy every day. Several studies reveal that global CO_2 emissions from the Information and Communication Technology sector are similar to or more than that of the entire aviation industry. With the advent of cryptocurrency and 5G, the numbers will only spiral up. Digital presence, especially in this binge-watching, working-from-home environment, consumes a considerable amount of energy that is often not talked about. Essentially the amount of CO_2 emissions from digital technology production and data transfer is the digital carbon footprint of an individual or an organisation.

Data centres that house all our data and have to be active at all times, every day consume a huge amount of electricity. Energy reserves and distribution systems, powerful internet connections coupled with generators and ventilation/cooling systems create a surge in demand for energy. Effectively, 40 percent of energy in the data centres is dedicated to the cooling systems. To put things into perspective, the internet per year consumes as much electricity as major countries such as Germany do!

The world is moving towards a cleaner source of energy away from fossil fuels. The policy documentation of countries has formulated plans for a greener planet through the use of renewable sources of energy. Sustainable growth is at the heart of climate conversations. Energy sources such as solar or wind might be the future, losing the long dependency on coal for everyday energy consumption. Renewable energy can diminish digital energy consumption by 80 percent as and when it arrives on a full scale.

So, the question remains, what can one do to reduce their carbon footprint till the renewable future of energy arrives. Several steps can help decrease the digital carbon footprint.

ENERGY EFFICIENT WEBSITE

A website can produce almost 2000 kg of CO_2 a year in emissions from page views. The more complex the website, the more the emission.

- **Page Design** – Compressed images, clever use of fonts and reduced usage of videos all contribute to a faster website loading time reducing page weight and making the website cleaner. A complex website might appear beautiful to the eye but is harmful to the planet.
- **SEO** – Search Engine Optimisation ranks websites according to a match with a keyword. A successful SEO helps websites appear faster on a search thus reducing time spent by the individual on the web finding information and cutting emissions.
- **Less JavaScript and a clean code** – Complex codes and heavy use of JavaScript (JS) increase the weight of a page and increase CPU usage for the website visitor. Efficient use of JS and simple codes can increase efficiency and decrease CO_2 emissions.
- **Accelerated Mobile Pages (AMP)** – AMP cuts down unnecessary file weight and codes. It displays a simple web page for mobile devices. If the original website is not lean, AMP can help with faster load times. However, every publisher